DATA-DRIVEN FINANCIAL RISK MANAGEMENT

SECTION 5: ORIGINATION ANALYTICS

Building the Application Scorecard — From Data to Credit Decision

A comprehensive 15-session course designed for Data Science students, transforming theoretical knowledge into practical skills.

15 SESSIONS TOTAL | ~ 60 SLIDES PER SESSION | THEORY TO PRACTICE FOCUS

Lecturer: Dr. Phạm Thị Nguyệt Nga

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RECALL

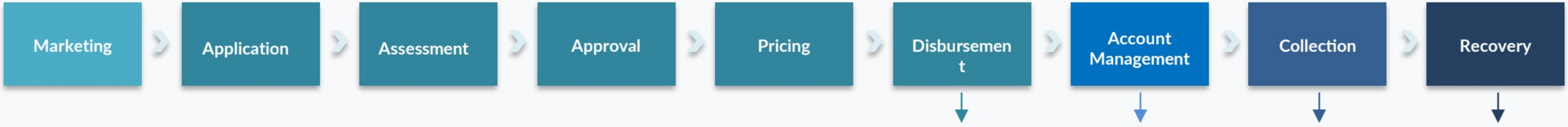
SECTION 4: CREDIT RISK MANAGEMENT

A Master Map for Credit Lifecycle

The Two-Layer Credit Lifecycle

Purpose — understand why loan processing is sequential, but risk management never stops.

LAYER 1 — LOAN LIFECYCLE



LAYER 2 — CONTINUOUS RISK CAPABILITIES (always on at bank level — a loan enters here at disbursement)

Risk Monitoring

Daily trigger tracking — all active loans

Portfolio Monitoring

Sector concentration, NPL, roll rate

Internal Reporting

Board, Risk Committee, ALCO

External Reporting

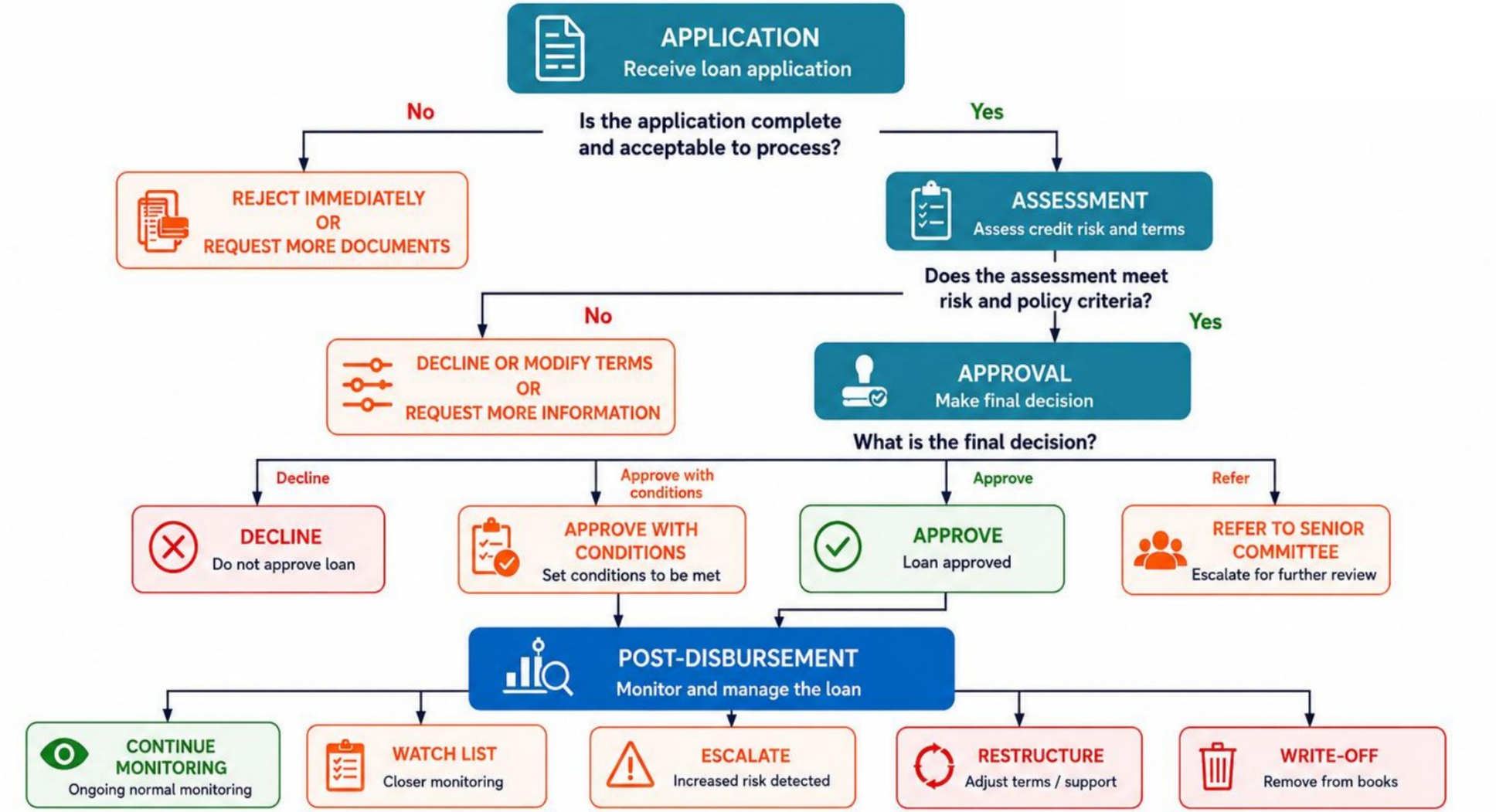
SBV and CIC — Circulars 15, 31, 41, 22

The loan has a beginning and an end

but

risk management never stops.

The Lifecycle Is Not a Straight Line



A loan is a DECISION TREE, not a conveyor belt.
Every branch is a **risk** decision. Every decision has a risk objective behind it.

Analytics Capability Map

Purpose — connect each business decision to the analytics capability that supports it. Full explanation in 4B.

Stage	Decision	Analytics Capability	Section
Marketing	Who to target?	Customer Segmentation	Section 4
Application	Identity valid?	KYC & Screening	Section 4
Assessment	**How risky?**	**Credit Scoring**	**Section 5**
Approval	**Approve?**	**Decision Engine**	**Section 5**
Pricing	**What rate?**	**Risk-Based Pricing**	**Section 5**
Account Management	Risk increasing?	Behaviour Scoring	Section 6
Account Management	Deterioration imminent?	Early Warning System	Section 6
Collection	Which strategy?	Collection Scoring	Section 7
Recovery	How much to recover?	Recovery Analytics	Section 7
Portfolio	Within appetite?	Portfolio Analytics	Section 8

Typical Score Types

Score Type	Target Population	Types of Information Used	No Times Score is Calculated	Prediction	Decisions Based on Scoring
Application	Customers applying for a **new loan / credit facility**	• Static information collected at moment of application (**application form**) • Internal Sources of information • External Sources of information	Once, at application point	Probability that a customer will reach severe delinquency status (e.g. 90+DPD) in the **post disbursal period*** (e.g. 12M)	• Accept / Reject / Refer • Loan Pricing • Limit Assignment
Behavioural	**Customers existing to bank** (have active accounts with the bank)	• Dynamic behavior information (**internal bank** customer/ transactional management systems) • External Sources of information	Potentially once a month or more often, when new dynamic summary information is available (as long as the account is active)	Probability that **a performing customer** will reach severe delinquency status (e.g. 30+DPD) in **the upcoming months*** (e.g. 3M)	• Price Review • Up-Sell / Cross-sell • Limit Review, Shadow Limit Management • Early Warning Strategy
Collection	**Customers existing to bank** (have active accounts with the bank)	• Dynamic behavior information (**internal bank** collection systems) • External Sources of information	Could be calculated daily depending on the collection strategy	Probability that a **delinquent customer** will reach worse delinquency status in the **upcoming months**	• Collection Effort Allocation • Collection Action • Transfer to Agency • Settlement / Re-structure / Close

Information Sources: Internal vs. External

Neither alone is sufficient for a complete risk picture

INTERNAL SOURCES — Bank controls these

Application data — borrower-declared
CBS transaction history — system-generated
Payment records — DPD, amounts, timing
Relationship history — 10 years with Company A
Collateral documentation — registered value
Covenant compliance log — LOS tracked

Richer in behavioral detail but bank-specific — no cross-bank view!

EXTERNAL SOURCES — Bank queries these

CIC — cross-bank total indebtedness (Cir. 15/2023)
Land registry — collateral title verification
Tax authority — revenue verification
Social insurance — income verification
Market/sector indices — Real Estate data (with Company A)
Business registry — legal entity status

Provides the cross-bank picture.
Essential for total indebtedness analysis.

Alternative Data: Bridging the Thin-File Problem

When traditional sources fail — first-time borrowers, SMEs, informal sector

Traditional sources fail when: first-time borrower | SME with limited financials | informal sector business | startup with no history

BANK TRANSACTION ANALYSIS

Salary credit regularity
Merchant category mix
Seasonal cash flow pattern

→ Proxy for income stability

TELCO DATA (selected markets)

Airtime top-up behavior
Bill payment regularity
Data package purchases

→ Proxy for financial discipline

E-COMMERCE (business borrowers)

Seller rating / review history
Order volume trend
Return rate / disputes

→ Proxy for business health

Proxies lift *Completeness*, not *Verified* — to use at Approval:
(1) **Statistical bias check** — proxy discrimination risk
(2) **Regulatory acceptability** — data consent rules
(3) **Temporal stability** — patterns may shift when alternative data platforms change

Source Reliability: A Working Framework

Must understand source limitations — not just use the data

Data Source	Coverage	Timeliness	Accuracy	Legal / Access Basis
Application Form	High	Real-time	Customer-declared — unverified	Internal policy
CIC Report	High	Batch	Bank-reported — collective quality	Mandatory — Cir. 15/2023
CBS Transactions	High	Real-time	System-generated — high reliability	Internal
Collateral Appraisal	Loan-specific	Periodic	Professional estimate — snapshot	Policy-governed
Tax Records	Medium	T-3 months	Government-verified — high accuracy	Voluntary / negotiated
Sector Indices	Medium	Monthly	Third-party — index-based	Internal risk use

Key risk: CIC is batch. An approval decision today uses credit data up to a month old. This is the most significant timing gap in the Application, hence Approval pipeline

DATA-DRIVEN FINANCIAL RISK MANAGEMENT

SECTION 5: ORIGINATION ANALYTICS

Building the Application Scorecard — From Data to Credit Decision

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Part A: Application Scorecard — Data → Features → Model → Score

Part B: Score → PD → ECL → Capital → RAROC → Credit Decision

From Credit Lifecycle to Credit Analytics

WHAT SECTION 4 GAVE US

- The PROCESS: lifecycle stages
- The DECISIONS: accept / monitor / collect
- The DATA CAPACITY: which capacity support which decision; an overview of important score types
- The DATA NEEDS: what info is required at each stage; information asymmetry; internal data vs external data; alternative data; importance of data quality
- The DATA ANALYTICS PROCESS

WHAT SECTION 5 ADDS

- The ANALYTICS ENGINE behind each decision
- APPLICATION SCORECARD: turning data into a credit score
- FEATURE ENGINEERING: WOE + IV (what predicts default?)
- Score → PD calibration (risk quantity)
- ECL → Capital → RAROC (the full decision chain)

In Section 4 we mapped the PROCESS. In Section 5 we open the ENGINE ROOM.

Three Models, Three Moments

Credit analytics does not stop at origination. There is a model for every major risk moment.

① APPLICATION SCORECARD

- Risk moment: ORIGINATION
- Question: Should we lend?
- Input: Application form + bureau data
- Output: Credit score (rank-order risk)
- Decision: Approve / Manual / Reject

★ SECTION 5 FOCUS

② BEHAVIOR SCORECARD

- Risk moment: ACCOUNT MANAGEMENT
- Question: How is this customer performing?
- Input: Transaction history + payment patterns
- Output: Behavior score (credit limit / repricing)
- Decision: Retain / Upgrade / Restrict

→ Section 6

③ EARLY WARNING MODEL

- Risk moment: MONITORING
- Question: Is this customer showing stress?
- Input: Transaction history + payment behavior + external signals
- Output: EWS flag (risk tier change)
- Decision: Contact / Restructure / Provision

→ Section 6

📌 *Note: Siddiqi (2017) treats Behavior Scorecard as one framework serving both account mgmt and monitoring purposes. In practice, banks typically build separate models — Industry Practice, consistent with Cir. 83/2025 Item 22.*

● SID ● VN-REG ● IP

What We Cover in Section 5

SECTION 5A

Building the Application Scorecard

- Block 1: Business Context — the origination decision
- Block 2: Risk Data Universe — what data to use
- Block 3: Feature Engineering — WOE and IV
- Block 4: Model Development — from regression to scorecard
- Block 5: Scorecard Output — score distribution, cutoffs



SECTION 5B

From Score to Decision

- Block 1: Decision Framework — score is not the decision
- Block 2: Quantification Chain — PD · EAD · LGD → ECL
- Block 3: Capital Gate — Credit VaR · EC · RAROC
- Block 4: Full Decision Chain — visual walkthrough
- Block 5: Closing — customer journey + preview Section 6

Block 1: Business Context — the origination decision

The Three-Layer Framework

Applied consistently across every model in this program:

LAYER 1
Business Decision

- What is the business question?
- Should we approve this loan?
- At what limit? At what rate?
- Who makes the final call?

Owner: Credit Officer / Policy

LAYER 2
Risk Analytics

- What does the data say?
- Score: rank-order risk
- PD: probability of default
- ECL = PD × EAD × LGD
- Credit VaR, Economic Capital

Owner: Risk Analytics / Quant

LAYER 3
Decision Policy

- What are the rules?
- Score cutoffs (Accept / Manual / Reject)
- RAROC ≥ cost of equity → Approve
- Override governance
- Regulatory compliance

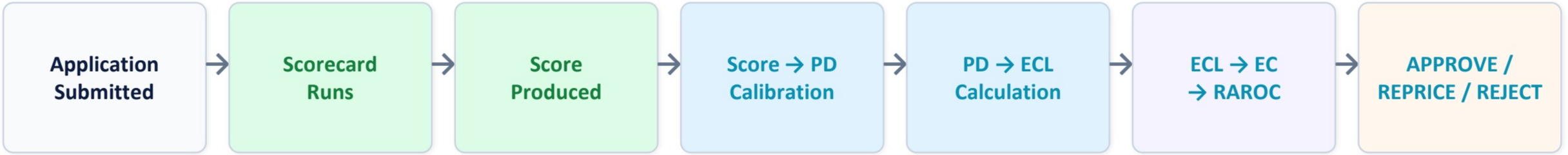
Owner: Risk Policy / CRO

This framework is the spine of the entire course. Every analytical tool we build sits in Layer 2 — feeding into Layer 1 (business) and constrained by Layer 3 (policy).

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One Customer Journey (Preview)

We will follow one retail loan applicant through the full analytics chain. We return to this slide at the end of Section 5B to close the arc.



SECTION 5A: Building the Scorecard

SECTION 5B: From Score to Decision

OUR ILLUSTRATIVE CUSTOMER:

Nguyen Van A — 34 years old · Salaried employee · Monthly income: VND 15M · Requests: personal loan VND 150M, 36 months · No existing bureau record (thin file)

We will return to this customer at the end of Section 5B — and trace every number from application to decision.

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The Business Decision at Origination

BLOCK 1: BUSINESS CONTEXT — Slide 1.1

One question drives everything at origination:

"Should we lend to this applicant — and if yes, on what terms?"

THREE-LAYER FRAMEWORK — Applied at Origination:

Layer 1 Business Decision	Layer 2 Risk Analytics	Layer 3 Decision Policy
Approve / manual review / reject	Application scorecard → credit score	Score cutoffs
Credit limit + interest rate	Score → PD estimate	RAROC gate
		Regulatory compliance

The Application Scorecard is the primary analytics engine in Layer 2 at the origination stage.

What Is an Application Scorecard?

BLOCK 1: BUSINESS CONTEXT — Slide 1.2

A statistical model that assigns a numerical score to each loan applicant, rank-ordering them by their likelihood of default. Higher score = lower risk.

WHAT IT IS:

- A rank-ordering tool — separates good from bad credit risks
- A consistent, automated decision aid (same rules, every applicant)
- Built from historical data: past applicants who were Good or Bad
- Output: a score (e.g. 600 points = odds 50:1) — higher = safer
- Anatomy: Characteristics → Attributes → Points → Total Score

WHAT IT IS NOT:

- Not a decision — the score is an input, not the output
- Not a prediction of individual default — it predicts relative risk
- Not permanent — scores drift; models need monitoring & recalibration
- Not a black box — regulatory explainability is required
- Not enough on its own — must be combined with policy (Layer 3)

📌 *In plain English: A score tells us who is riskier than whom — not whether one person will definitely default.*

Model Outcome – Example

Case 1 — Rank-ordering, STRONG separation good from bad credit risks

High risk ↑ Low risk

Score Range	Risk Tranche	Bad	Good	Indeterminate	Exclusion	Total	Bad Rate	Total %	Cumulative % of Bads	Multiplier
low - 519	A	11,714	2,686	1,464	655	16,519	81.3%	9.9%	23.2%	**19.86**
519 <- 539	B	8,781	5,398	2,561	454	17,194	61.9%	10.3%	40.6%	
539 <- 554	C	7,704	6,880	2,592	372	17,548	52.8%	10.6%	55.9%	
554 <- 565	D	5,971	7,596	2,738	502	16,807	44.0%	10.1%	67.8%	
565 <- 577	E	5,429	9,897	3,047	437	18,810	35.4%	11.3%	78.5%	
577 <- 596	F	3,483	8,884	2,206	220	14,793	28.2%	8.9%	85.4%	
596 <- 607	G	3,719	12,111	2,827	254	18,911	23.5%	11.4%	92.8%	
607 <- 650	H	1,941	11,122	1,732	204	14,999	14.9%	9.0%	96.7%	
650 <- 686	I	1,106	13,028	1,290	196	15,620	7.8%	9.4%	98.9%	
686 <- high	J	578	13,533	842	149	15,102	4.1%	9.1%	100.0%	
Total		**50,426**	**91,135**	**21,299**	**3,443**	**166,303**	**35.6%**	**100.0%**		

Case 2 — Rank-ordering, WEAK separation good from bad credit risks

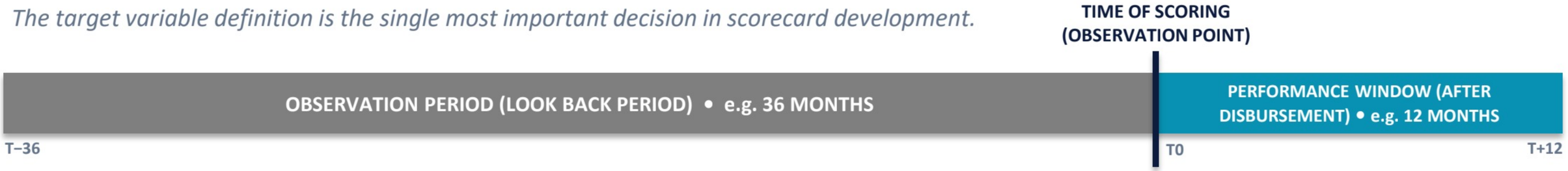
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519 <- 539	B	6,111	8,068	2,561	454	17,194	43.1%	10.3%	25.0%	
539 <- 554	C	5,974	8,610	2,592	372	17,548	41.0%	10.6%	36.9%	
554 <- 565	D	5,267	8,300	2,738	502	16,807	38.8%	10.1%	47.3%	
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Defining Good, Bad, Indeterminate, and Exclusion

BLOCK 1: BUSINESS CONTEXT — Slide 1.3

The target variable definition is the single most important decision in scorecard development.



GOOD

Clean performance throughout the performance window.
Example:

- Always Group 1 and not being charge-off, write-off, or bankrupt during the performance window;
- New to credit or Group 1 (bureau level) at the time of scoring

BAD

Reached the bad definition during the performance window.
Example:

- Ever being 91^{+} days past due or Group 3-5 (most common), or charge-off, write-off, bankrupt during the performance window

INDETERMINATE & EXCLUSION

- **Exclusion:** Insufficient performance(e.g. less than 6 monthly payments); deceased, fraud, staff/VIP accounts, policy declines etc.
- **Indeterminate:** sufficient performance but not classified as Good or Bad (e.g. ever 10–90 DPD (Group 2) but never 91+)
- **Usually excluded from development sample** to avoid contamination.

WHY THE 'BAD' DEFINITION DRIVES EVERYTHING:

- Changes the bad definition → changes which accounts are labeled 'bad' → changes the model entirely
- Performance window for Application Scorecard: varies by product (typically 12–24 months); determined by vintage analysis
- Bad rate in development sample directly affects model calibration and cutoff setting

Block 2: Risk Data Universe — what data to use

The Risk Data Universe

Four data layers feed the Application Scorecard.

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① INTERNAL DATA

Application + Transaction

- **Demographic**: age, marital status, dependents;
- **Employment:** declared income, stability, duration, industry
- **Residential**: area, type, time at address, home ownership;
- **Applied loan:** amount, tenor, purpose, collateral
- **Customer relationship:** relationship depth, products held
- **Transaction & spending patterns**, account balance
- **Account Activities:** login frequency, time, contact behavior

② BUREAU DATA

Credit Reference Agency

- **Credit information:** #institutions in relationship, outstanding balance, #inquiries, credit card usage
- **Debt group history:** current debt group, worst group in last 12/24/36 months
- **Bureau score** (from CIC, PCB, Nice)

③ FINANCIAL DATA

Cross-check Income & Cash Flow

- **Bank statements** (3–12 months)
- **Salary slips**
- **Tax / Social insurance** contributions
- **Other income:** rental, business (with proof; usually haircut)
- **Verified assets:** property, vehicle, savings

Derived Debt-to-income ratio (CIC obligations / verified income)

④ ALTERNATIVE DATA

Digital & Behavioral

- **Payment behavior:** Utility bill payment history (electricity, water, internet); E-wallet transactions & top-up patterns
- **Third-party scores** (*): **Telco score**: billing, top-up, usage (e.g. via Data Nest); **E-commerce** / BNPL: shopping & repayment behavior (e.g. Shopee SPayLater**); Platform trust score** (e.g. Zalopay trust score)
- **Device & digital footprint** (IP): Device type, telco provider, app usage; **Application behavior:** time to complete form, number of edits; **Geolocation** (sensitive: needs explicit consent)

⚠️ **Use only data available up to T0** (no post-application information), **collected with customer consent**.
⚠️ Every feature **must pass compliance review**; selection is then **driven by predictive power and stability**.

(*) Third-party scores: limited explainability; customer consent required.

Alternative Data: The Caveats

Siddiqi (2017) is explicit: alternative data requires four risk assessments before use.

① PRIVACY LAWS

Using customer data without proper consent or legal basis violates privacy law. In Vietnam: Personal Data Protection Law 2025 and Data Law 2024. Consent must be explicit, documented, and specific to the credit scoring purpose.

② REPUTATIONAL RISK

Customers may react negatively if they learn their social media or phone data was used to deny their loan application. Media coverage of such practices has forced banks to withdraw data sources.

③ RELIABILITY

Alternative data is often self-created or self-reported and may be gameable. Social media profiles can be fabricated. Telecom top-up behavior may not be available continuously. Can this data source support a model in production for 5+ years?

④ REGULATORY DEPENDENCY

Rules for alternative data change. A data source that is permitted today may be prohibited next year. The model must not depend on data sources that have an uncertain regulatory future => **A modular design** (core model + separate alternative-data module) lets the bank drop a data source without rebuilding the whole scorecard.

⚠️ *Note: Device fingerprint, GPS geolocation, in-app behavior = Industry Practice (IP) — NOT coved in Siddiqi (2017).*
Telecom data = mentioned as proxy discrimination RISK, not a feature recommendation.

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Data Privacy Compliance: Risk Assessment by Feature

Feature Group	Privacy Category	Risk Level	Key Concern	Flag
GPS Geolocation	Sensitive personal data	🔴 HIGH	Personal Data Protection Law 2025: location = sensitive; Siddiqi warns: cell-tower data → discrimination risk	SID+VN
Social Connections (Facebook/Twitter)	Personal data + reputation	🔴 HIGH	Social network ≠ creditworthiness; Financial Institution Law: customer info confidentiality; reputational risk	SID+VN
Telecom Data	Behavioral / sensitive	🔴 HIGH	Siddiqi explicitly cites telco as PROXY DISCRIMINATION risk — not a feature recommendation	SID+VN
Device Fingerprint	Behavioral / proxy	🟡 MEDIUM	Handset type → proxy for income/ethnicity → Fair Lending violation risk	IP
In-App Behavior (screen time / taps)	Behavioral / consent	🟡 MEDIUM	Requires explicit consent per Data Law 2024; gameable; reliability concern	SID+VN
Utility Bills	Non-sensitive	🟢 LOW	Payment discipline proxy; acceptable with consent; Siddiqi pp.138-139 supports	SID
Bureau + Application	Standard credit data	🟢 LOW	Core regulated data; clear legal basis; required consent at application	SID+VN

→ Principle: Obtain written/electronic consent for ALL alternative data; remove all protected attributes (gender, marital status, religion, nationality).

● SID ● VN-REG ● IP

Data Compliance Checklist: "Safe to Model?"

Apply these 6 questions to every variable BEFORE it enters the model:

1 Privacy & Consent

Has the customer **explicitly consented** to using this data for credit scoring? Is it sensitive personal data (location, telecom, social)? (Law on Personal Data Protection 2025 — Law 91/2025, Art. 27.1(b))

2 Protected Attributes

Does this variable **violate Fair Lending**? (gender, religion, marital status, nationality, ethnicity — never allowed)

3 Proxy Discrimination

Could this variable be **an indirect proxy for a protected group**? (zip code → income/ethnicity; device type → social class)

4 Causality & Reputation

Is the relationship between this variable and credit risk **logically explainable**? Could using this variable lead to public criticism or **concerns about fairness**?

5 Source Reliability

Can this data source **be gamed by applicants**? Is it guaranteed to be continuously available for 5+ years of model life?

6 Explainability

If a regulator or rejected customer asks WHY **— can the model explain** this variable's contribution? (Required: AI Law 2025)

📌 Teaching note: "If you cannot answer YES to all six — the variable stays out."
Discussion: If a variable has very high IV but violates customer privacy — do you sacrifice model accuracy, or accept legal and model governance risk?

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Block 3: The Scorecard Development Pipeline

HOW TO DO EFFECTIVE DATA ANALYTICS?

END-TO-END ANALYTICS PROCESS — 7 KEY STEPS

BUSINESS UNDERSTANDING

Define Business Objectives

Identify key business problems or goals; align with overall strategic priorities.

E.g: Which type of customer (individuals / corporates; new to credit / existing customer); Type of prediction (90+PPD in next 12M); portfolio size (cumulative # customers, # NPL customers); Historical understanding of model development, model quality & data sources; Measure of success (NPL rate); Current Risk Pricing; Key Stake holders

Problem Analysis

Apply structured analytical frameworks to identify KPIs, expectations, assumptions, and relationships.

E.g: Define prediction target; Interviewing our experts to understand what type of information are essential for predicting customer delinquency? What metrics? How these help predicting defaults? What is underlying assumptions (e.g. more revenue growth means less risk)? Important levels? Expected correlation between metrics ect.

Design & Plan

Identify data sources; assess availability, quality, and relevance for the analysis.



Share, Monitor & Measure

Share results safely; automate reporting; continuously monitor and measure effectiveness.

Interpret & Suggest Actions

Convert analytical findings into clear, actionable recommendations for stakeholders.

Explore & Analyze

Use statistical and advanced techniques to discover insights, patterns, and relationships.

Collect, Process & Organize Data

Gather relevant data from multiple sources; validate completeness and accuracy.
Check quality, clean and transform data into a structured, analysis-ready format.

VALUE

ACTION

INSIGHT

INFORMATION

DATA

The Scorecard Development Pipeline

Phase A: Data Preparation | **Phase B: Feature Engineering** | **Phase C: Model Build & Validate**

Define Target

1 **Performance window & observation period**

2 **Good / Bad definition**

- Segmentation basis (for uniform risk profiling)

OUTPUT

Target & Segmentation Doc

- Roll rate & vintage
- Segment profiles
- Parameter choices

Data Discovery

- Collect data
- Data quality checks
- Data cleansing

3 **Sample data**

4 **Dev / Val split**

OUTPUT

Data Quality Report

- Missing %, outliers
- Data distributions
- Data treatment & summary

Feature Engineering

- Variable creation

5 **Binning, WOE, IV**

- Correlation checks
- Business logic check

6 **Feature selection**

OUTPUT

Data Analysis Doc

- Variable waterfall
- Logical filtering
- Reasons for exclusion
- Results of analysis

Develop Model

7 **KGB scorecard**

8 **Reject inference → AGB**

- Final scorecard on AGB
- AI/ML challenger (IP)

OUTPUT

Model Performance Report

- Regression assumptions & results (GINI, KS)
- ML challenger tuning

Validate & Document

9 **Validation & approval**

- Out-of-sample & out-of-time tests
- Model Risk sign-off

OUTPUT

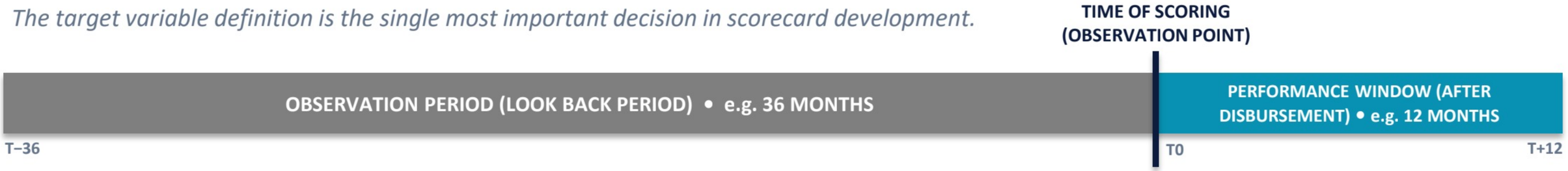
Validation Report

- PSI, CSI (HHI = IP)
- Out-of-time results
- KPI thresholds & full documentation

Iterative: steps 5–7 repeat until the scorecard is stable and makes business sense. **Step 9** needs independent sign-off from Model Risk, not only the model builders.

Recall - Defining Good, Bad, Indeterminate, and Exclusion

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Block 4: Data Discovery

The Scorecard Development Pipeline

Phase A: Data Preparation

Define Target

1. **Performance window & observation period**
2. **Good / Bad definition**
- Segmentation basis (for uniform risk profiling)

OUTPUT

Target & Segmentation Doc

- Roll rate & vintage
- Segment profiles
- Parameter choices

Data Discovery

- Collect data
- Data quality checks
- Data cleansing

3. **Sample data**
4. **Dev / Val split**

OUTPUT

Data Quality Report

- Missing %, outliers
- Data distributions
- Data treatment & summary

Iterative: steps 5–7 repeat until the scorecard is stable and makes business sense. **Step 9** needs independent sign-off from Model Risk, not only the model builders.

The Scorecard Development Pipeline

Data Discovery

- Collect data
- Data quality checks
- Data cleansing

3 **Sample data**

4 **Dev / Val split**

OUTPUT

Data Quality Report

- Missing %, outliers
- Data distributions
- Data treatment & summary

Data handling

(★ = standard practice)

Method	Pros	Cons	Best when
MISSING VALUES			
Separate "Missing" bin ★	Keeps the signal; the bin gets its own risk score	If <5% of records, merge with the bin of closest bad rate	Missing carries a risk signal or is structural (no CIC record, never delinquent)
Impute (median / mode)	Simple; keeps all records	Hides the signal; distorts distribution	Missing is random (no risk signal), e.g. system errors
Drop records / variable	Clean development data	Loses data; biased if not random	Few random records (e.g. <5%); Drop the variable if >30% missing for no structural or predictive reason
OUTLIERS			
Coarse binning ★	Extremes go to the end bins	Less detail in the tails	Default for binned-variable scorecards
Cap & floor (P1 / P99)	Keeps records; limits extremes	Cut-off is a judgment call	Raw continuous inputs (ML models, ratios such as DTI)
Verify & correct errors	Fixes impossible values	Manual; needs source documents	Impossible values (e.g. age 150)

Null ≠ 0, and "not applicable" ≠ "unknown". Give it its own bin or a substitute variable; never fill with 0.

Example:

- **No CIC record (new to credit)**: put these in a "No bureau history" bin, kept apart from Group 1. Their risk differs from customers with a clean credit history.
- **Never delinquent**: "months since last delinquency" will be blank. That blank means the best bin, so it must not be imputed or set to 0 (0 = delinquent this month).
- **Near-zero denominator** (DTI with no income, growth from ~0 balance): assign a special bin before computing the ratio; capping would mix these with genuinely high DTI.

Coarse Binning: Monthly Income Example

Step 1: 9 fine bins → Step 2: merge into 4 coarse bins + Missing

Fine bin (income)	% accts	Bad rate	Coarse bin	Bad rate
0–4M	4.7%	11.4%		
5–7M	11.8%	7.1%	**≤ 9M**	**7.3%**
8–9M	11.1%	5.7%		
10–12M	15.6%	5.3%		
13–15M	10.8%	4.6%	**10–19M**	**4.7%**
16–19M	11.8%	3.9%		
20–29M	16.9%	3.3%		
30–39M	6.8%	2.8%	**20–39M**	**3.1%**
≥ 40M	6.0%	2.0%	**≥ 40M**	**2.0%**
Missing	4.4%	8.8%	**Missing**	**8.8%**

Song Hong Bank, training sample: 32,090 loans, 5.0% bad (Excel 2_KGB_WOE_IV).

Why merge: remove noise

Inside 10–19M the fine bins (5.3%, 4.6%, 3.9%) differ by about one point each — mostly noise. Coarse bins keep a clean downward trend and ≥ 5% of loans each (p.183).

Outliers: no extra rule needed

60 incomes above 500M (e.g. 2,470M, a unit error) fall in "≥ 40M" with a 2.0% bad rate, so they get the same score as 45M. They cannot distort the model (p.178).

Missing: its own bin

Missing income has an 8.8% bad rate, 1.75× the average. Imputing the median (14M) would put it in the 10–19M bin at 4.7% and understate its risk.

Trade-off: steps at boundaries

19M and 20M get different scores; 45M and 2,470M get the same. Detail inside each bin is lost.

Sample Design: Development vs. Validation

DEVELOPMENT SAMPLE

70 – 80% of total sample

- Used to fit the (logistic regression) model
- WOE transformation computed on this sample
- **Coefficients estimated from this data**
- **Minimum:** 2,000 Goods + **2,000 Bads** (Siddiqi).

With Vietnamese practice: bad rate of 10%-20% is sometime acceptable.

- **Statistically representative** of target population

VALIDATION (HOLD-OUT) SAMPLE

20 – 30% of total sample

- **Never seen by the model during development**
- Tests out-of-sample performance (generalizability)
- Model performance (KS, Gini, AUC) measured on BOTH samples
- Performance gap dev vs. val = sign of overfitting

SMALL SAMPLE SOLUTIONS

- Building model on a small sample can lead to **unreliable performance estimates** and **unstable variable selection**
- **K-fold cross-validation** and **bootstrap resampling** can help reduce this risk.
- However, performance estimates will still carry greater uncertainty than those based on a larger sample.

CLASS IMBALANCE (next slide)

- Typical bad rate: 2–8% in consumer lending
- **Imbalanced data → model biased toward Goods**
- Solution: **oversampling** (next slide)

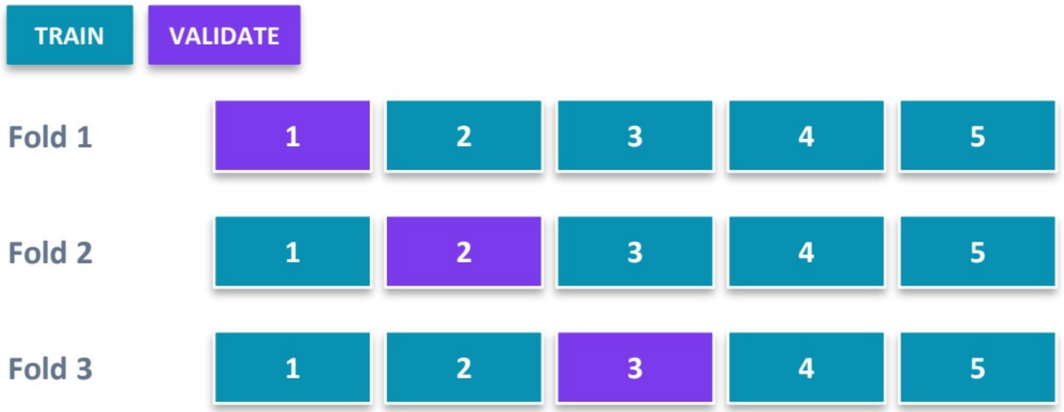
Cross-validation: Would we reach the same conclusion with a different data split?

Evaluate the model across multiple data splits to avoid conclusions that depend on a particularly favorable or unfavorable split.

1 ROTATE THE VALIDATION FOLD

Train on k−1 folds; validate on the remaining fold

Repeat until every observation has been validated exactly once.



Best for: model comparison and hyperparameter selection

Process: Aggregate results → Select configuration → Refit final model

2 AGGREGATE EVERY FOLD — NEVER PICK THE BEST ONE

MEAN PERFORMANCE	Is the model strong on average?
VARIATION ACROSS FOLDS	Is the result stable across splits?

TRUSTWORTHY — Strong mean + low variation

42%	40%	44%	41%	43%	**Mean 42%**

NOT TRUSTWORTHY — Same mean + wide variation

25%	60%	31%	55%	39%	**Mean 42%**

DECISION: choose high mean + low variation → REFIT: train the selected configuration on the full development sample.

• SID

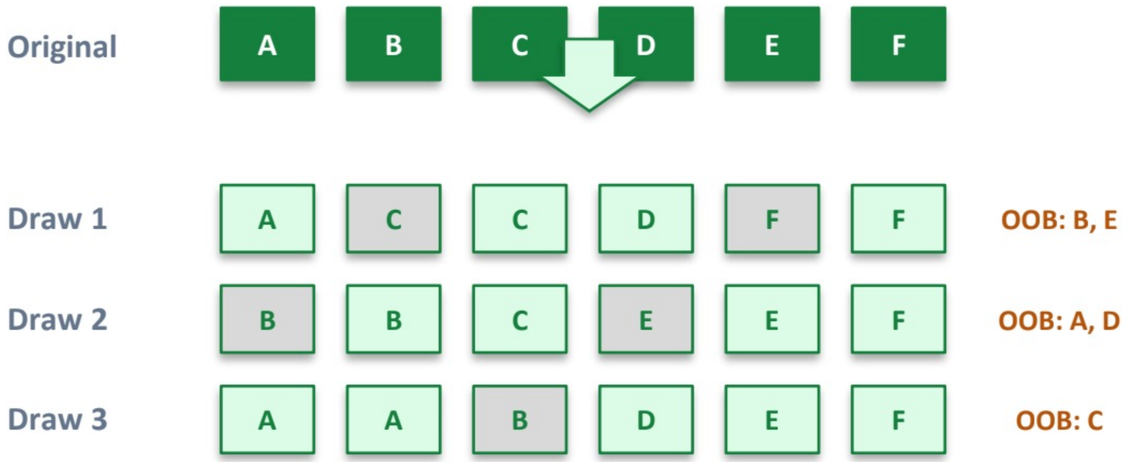
Bootstrap: How Much Would Results Change with a Different Sample?

Resample with replacement to estimate how model results may vary across possible samples from the same population.

1 SAMPLE WITH REPLACEMENT

Repeatedly draw n observations with replacement

Duplicates may appear; observations not drawn form the out-of-bag sample.



Best for: uncertainty, confidence intervals and stability

Process: Aggregate draws → Quantify uncertainty → Refit final model

2 AGGREGATE ALL DRAWS — NEVER PICK THE BEST ONE

UNCERTAINTY How wide is the performance distribution?

STABILITY Do variables and coefficients remain consistent?

ILLUSTRATIVE GINI ACROSS DRAWS

TRUSTWORTHY Narrow distribution

40%	41%	42%	43%	44%	Mean 42%

HIGH UNCERTAINTY Same mean, wide distribution

25%	35%	42%	50%	58%	Mean 42%

DECISION: trust narrow uncertainty + stable variables → REFIT: train the selected specification on the full development sample.

• SID

Handling Imbalanced Data: Oversampling

In consumer lending, bad rates are typically 2–8%. This extreme class imbalance requires adjustment.

THE PROBLEM

- If 3% of accounts are Bad → 97% are Good
- A model that predicts 'Good' for everyone is 97% accurate — but completely useless
- **The model never 'sees' enough Bads to learn their patterns**
- Solution: artificially balance the sample before training

PRIOR PROBABILITY ADJUSTMENT

- If development bad rate = 30% but true rate = 3%, all predicted PDs must be scaled down
- Offset = ln(bad_rate_dev / bad_rate_true) — added to the model intercept

TWO APPROACHES:

① Oversampling / SMOTE

- **Replicate Bad accounts** to achieve 30–50% bad rate in development sample
- After building model, **adjust predicted probabilities** back to true bad rate (prior adjustment)

② Sampling Weights

- Weight each Bad observation more heavily during model training
- Mathematically equivalent to oversampling; may be preferred in regularized models

📌 Why this matters for calibration: oversampling inflates the apparent bad rate in the model. If you use oversampled predictions as PD estimates without adjustment, your ECL calculation will be seriously overstated. **Always re-calibrate to the true population bad rate** before using scores as PDs.

The Scorecard Development Pipeline

Phase A: Data Preparation | **Phase B: Feature Engineering**

Define Target

1. **Performance window & observation period**
2. **Good / Bad definition**
- Segmentation basis (for uniform risk profiling)

OUTPUT

Target & Segmentation Doc

- Roll rate & vintage
- Segment profiles
- Parameter choices

Data Discovery

- Collect data
- Data quality checks
- Data cleansing

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OUTPUT

Data Quality Report

- Missing %, outliers
- Data distributions
- Data treatment & summary

Feature Engineering

- Variable creation

5. **Binning, WOE, IV**

- Correlation checks
- Business logic check

6. **Feature selection**

OUTPUT

Data Analysis Doc

- Variable waterfall
- Logical filtering
- Reasons for exclusion
- Results of analysis

Block 5: Feature Engineering

WOE and IV & Feature Selection

FEATURE ENGINEERING

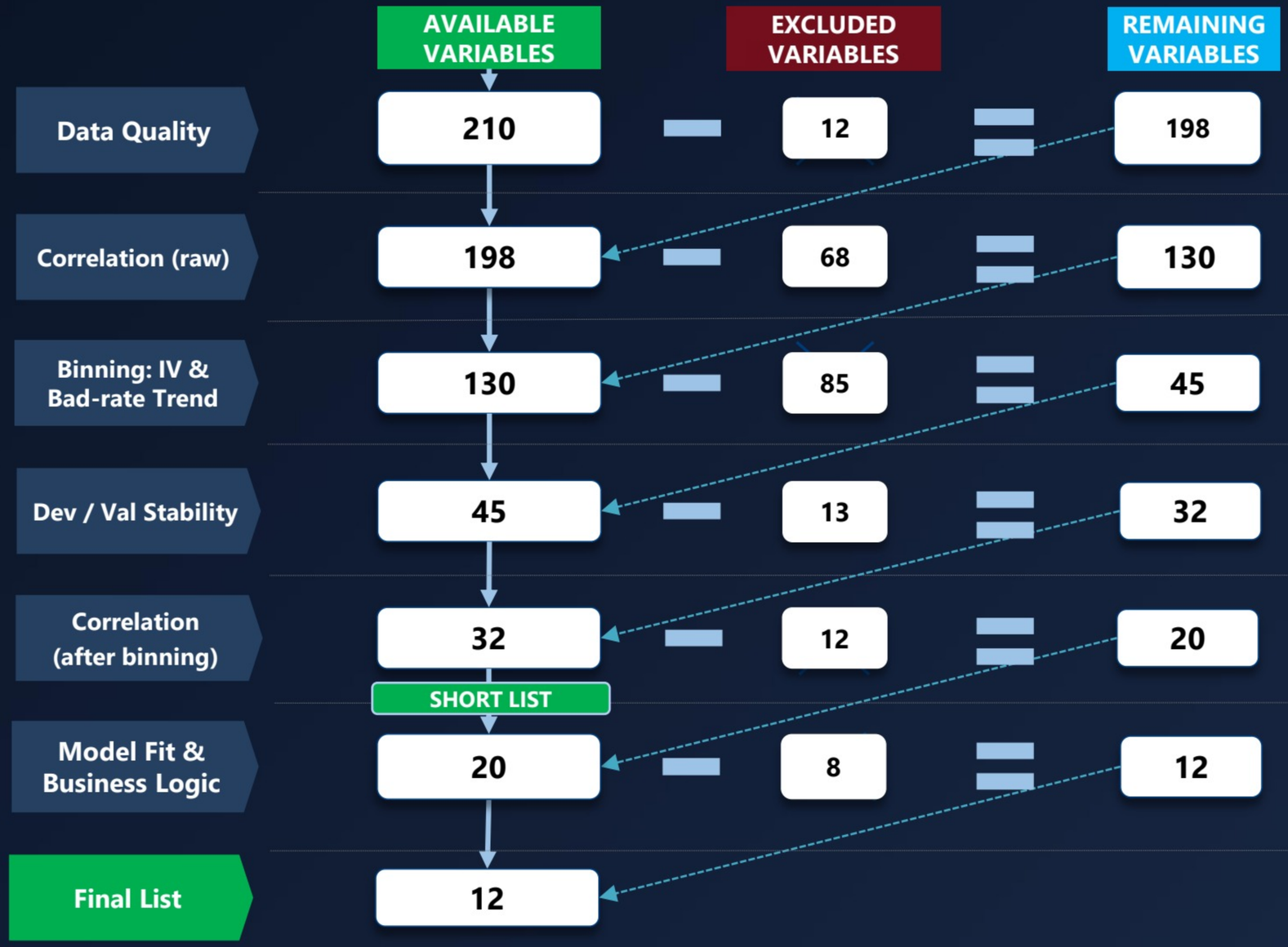
Attribute Transformation

- Variable grouping – ensure linearity, stability and interpretability
- Weight of Evidence (WOE) transformation
- Feature encoding (for ML algorithms)

Feature Selection

- Basis data quality and feasibility
- Linearity for conventional methodologies
- Predictive power (IV)
- Correlation and multicollinearity
- Business coherence/interpretability
- Model training

FEATURE REDUCTION



Purpose · Filter · Example

Purpose: Drop variables that cannot be used reliably
Filter: missing >30% (non-structural), constant value, not available at T0
e.g. Spouse income (80% blank), Mobile app logins (new field, not at T0)

Purpose: Remove duplicate measures of the same concept
Filter: |r| > 0.8 on raw values
e.g. Total CIC outstanding vs Total monthly CIC instalment

Purpose: Keep variables that predict on their own and make sense
Filter: IV < 0.02 (weak); bad-rate trend not monotonic or illogical
e.g. Number of dependents (flat bad rate), Time at employer (zig-zag)

Purpose: Keep variables that hold out of sample
Filter: Bad-rate trend reverses on validation; IV falls sharply
e.g. Months at current address, Card utilisation 3m

Purpose: Avoid multicollinearity among model inputs
Filter: |r| > 0.6 after binning or VIF > 5; keep the higher-IV variable
e.g. DTI vs Monthly income; # CIC lenders vs # active loans

Purpose: Keep variables that still add information alongside the others
Filter: p-value > 0.05; sign against bad-rate trend; fails business review (fairness, manipulable, not available in production)
e.g. Loan tenor (p = 0.21); Income (sign flips); Branch code (region proxy - which raises fairness concerns)

Final list (12): variables that pass every filter enter the scorecard, within the 10–20 industry range

Industry practice for logistic scorecards (banking industry): about 10–20 final variables.

From Raw Data to Predictive Features

Raw data is not model input. Feature engineering transforms data into risk signals.

RAW DATA

As collected

- income: 15,000,000
- employment: 'salaried'
- bureau: No bureau record
- zip: 70000
- age: 34

FEATURE ENGINEERING

Transform + encode

- Binning & grouping
- WOE transformation
- Missing value treatment
- Outliers absorbed by the end bins

MODEL INPUT

Predictive features

- WOE(dpd_cnt_12m)
- WOE(card_util_pct)
- WOE(cic_hist_months)
- WOE(income_mil)
- + 7 more (11 inputs)

- Domain expertise matters: a data scientist without credit knowledge may engineer the wrong features
- Feature engineering can be as impactful as model choice: a good WOE transform often beats a raw ML feature in credit scoring

Weight of Evidence (WOE) and Information Value (IV)

WEIGHT OF EVIDENCE (WOE)

$$WOE(i) = \ln [\ \%\ Goods(i) / \%\ Bads(i)\ ]$$

- Converts each attribute bin into a consistent risk scale
- Positive WOE → more Goods than Bads (lower risk bin)
- Negative WOE → more Bads than Goods (higher risk bin)
- WOE = 0 → bin has same Good/Bad rate as overall portfolio

INFORMATION VALUE (IV)

$$IV = \Sigma_i\ (\%\ Goods(i) - \%\ Bads(i)) \times WOE(i)$$

IV Value	Predictive Power
< 0.02	Useless — exclude
0.02 – 0.10	Weak
0.10 – 0.30	Medium — useful
0.30 – 0.50	Strong
> 0.50	Very strong — check for over-predicting (p.185)

EXAMPLE — Time in job, Song Hong training sample (32,090 loans):

Time in job	Bin	Bad	Good	% Bad	% Good	WOE	IV
≤ 11 months	G1	316	3,402	19.5%	11.2%	(0.56)	0.05
12–35 months	G2	560	10,351	34.6%	34.0%	(0.02)	0.00
36–83 months	G3	327	8,463	20.2%	27.8%	0.32	0.02
≥ 84 months	G4	124	5,220	7.7%	17.1%	0.80	0.08
Missing	G5	290	3,037	17.9%	10.0%	(0.59)	0.05
Total		**1,617**	**30,473**	**100%**	**100%**		**0.19**

Block 6: Model Development — from regression to scorecard

The Scorecard Development Pipeline

Phase A: Data Preparation | **Phase B: Feature Engineering** | **Phase C: Model Build & Validate**

Define Target

1 **Performance window & observation period**

2 **Good / Bad definition**

- Segmentation basis (for uniform risk profiling)

OUTPUT

Target & Segmentation Doc

- Roll rate & vintage
- Segment profiles
- Parameter choices

Data Discovery

- Collect data
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- Data cleansing

3 **Sample data**

4 **Dev / Val split**

OUTPUT

Data Quality Report

- Missing %, outliers
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- Data treatment & summary

Feature Engineering

- Variable creation

5 **Binning, WOE, IV**

- Correlation checks
- Business logic check

6 **Feature selection**

OUTPUT

Data Analysis Doc

- Variable waterfall
- Logical filtering
- Reasons for exclusion
- Results of analysis

Develop Model

7 **KGB scorecard**

8 **Reject inference → AGB**

- Final scorecard on AGB
- AI/ML challenger (IP)

OUTPUT

Model Performance Report

- Regression assumptions & results (GINI, KS)
- ML challenger tuning

Building the Model: Logistic Regression

Why logistic regression — not neural networks, not random forests?

WHY LOGISTIC REGRESSION

- Interpretable: each coefficient = direction + magnitude of risk
- Regulatory acceptance: can explain every score to regulators
- WOE inputs already capture non-linearity — no need for complex models
- Monotonic: higher WOE → consistently higher/lower score
- Stable: well-understood behavior over time

Model Performance Metrics

BLOCK 4: MODEL DEVELOPMENT — Slide 4.5

Three standard metrics to evaluate discriminatory power of a credit scorecard:

KS STATISTIC (Kolmogorov-Smirnov)

KS = max | F_Good(x) − F_Bad(x) |

Maximum separation between Good and Bad cumulative distributions. Measures how well the model separates the two populations at the best possible cutoff.

Benchmark:
KS > 40 = strong | 30–40 = acceptable | < 20 = weak

GINI COEFFICIENT

Gini = 2 × AUC − 1

Measures overall rank-ordering ability. Gini of 0 = random model. Gini of 100% = perfect separation. Derived from the Lorenz / Gains Curve.

Benchmark:
Gini > 60% = excellent | 45–60% = good | 35–45% = acceptable | < 35% = weak

c-STATISTIC / AUC (Area Under ROC Curve)

AUC = P(score_Good > score_Bad)

Probability that a randomly selected Good account scores higher than a randomly selected Bad account. AUC = 0.5 is random; AUC = 1.0 is perfect.

Link to Gini: AUC = (1 + Gini) / 2, e.g. Gini 50% = AUC 0.75.

Benchmark:
AUC > 0.80 = excellent | 0.725–0.80 = good | 0.675–0.725 = acceptable | < 0.675 = weak
Same cut-offs as Gini 60% / 45% / 35%

📌 *These three metrics measure the SAME underlying property (rank-order discrimination) in different ways. KS is most commonly used in practice.*

Model Performance Metrics – GINI

Index Gini measures the degree of improvement for the scorecard over a random decision-making system. Specifically, it measures the quotient of Area B (representing the efficiency of the score decision making) to Areas A plus B (representing the random decision making) as outlined below. The formula of Gini Index is:

$$\frac{Area\ B}{Area\ A + Area\ B} = 100 * \left\{ 1 - 2 * \sum_{i}^{n} \left[\left(1 - \sum_{j}^{i} \frac{b(x_j)}{T_b} \right) * \frac{g(x_j)}{T_g} \right] \right\}$$

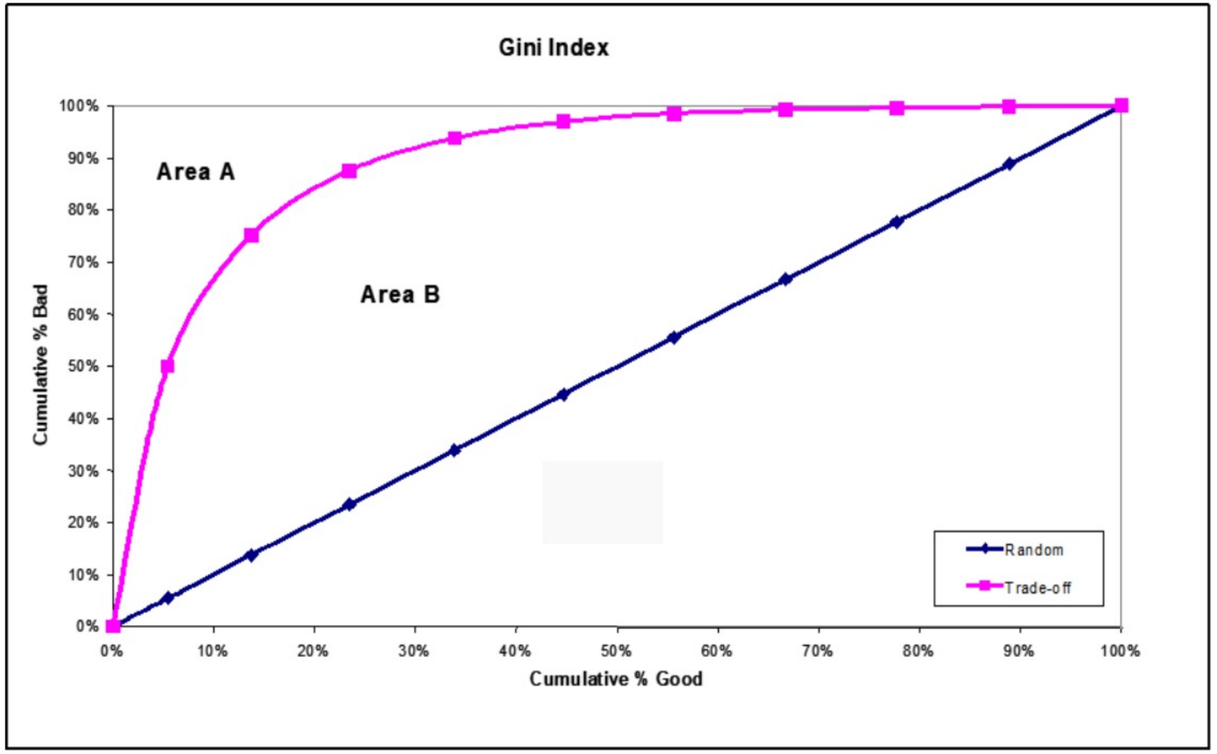
Where:

$b(x_j)$ =number of bads within the score band j

$g(x_j)$ =number of goods within the score band j

T_b = total number of bads

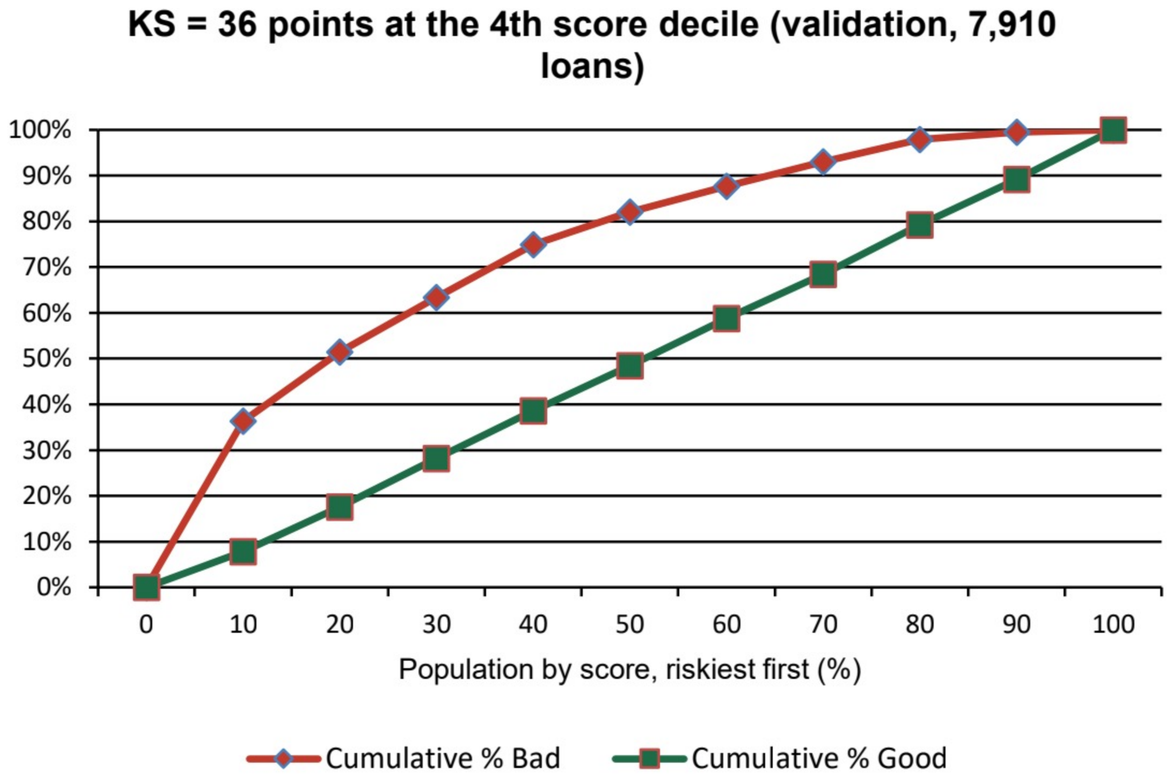
T_g = total number of goods



Model Performance Metrics – KS

The K-S statistic measures the maximum difference between the cumulative distribution of Bads and of Goods by score (p.253). In the Song Hong validation sample, 74.9% of the bad accounts and 37.6% of the good accounts have a score of 560 or lower. The difference is 74.9 − 37.6 = 37.3. No other cut-off gives a larger gap (37.4 at most), so KS ≈ 37.

KS ranges from 0 (no separation) to 100 (perfect separation). Application scorecards typically show 40–60; KS is measured at one point only, so the c-statistic is better for the whole range (p.251, 253–254).



Model Performance Metrics

Song Hong final scorecard — validation (holdout) sample, 2023: rank-ordering by score decile

Score range	Risk tranche	Bad	Good	Bad rate	Total %	Cum % of Bads	Cum % of Goods	Gini (trapezoid)	KS (Cum Bad − Cum Good)
low – 522	A	139	588	19.1%	9.2%	36.3%	7.8%	0.03	28.5%
523 – 539	B	58	739	7.3%	10.1%	51.4%	17.6%	0.09	33.8%
540 – 551	C	46	795	5.5%	10.6%	63.4%	28.2%	0.12	35.3%
552 – 561	D	44	784	5.3%	10.5%	74.9%	38.6%	0.14	36.3%
562 – 569	E	27	737	3.5%	9.7%	82.0%	48.4%	0.15	33.6%
570 – 577	F	22	785	2.7%	10.2%	87.7%	58.8%	0.18	28.9%
578 – 585	G	20	718	2.7%	9.3%	93.0%	68.4%	0.17	24.6%
586 – 595	H	19	813	2.3%	10.5%	97.9%	79.2%	0.21	18.7%
596 – 607	I	6	751	0.8%	9.6%	99.5%	89.2%	0.20	10.3%
608 – high	J	2	817	0.2%	10.4%	100.0%	100.0%	0.22	0.0%
Total		**383**	**7,527**	**4.8%**	**100.0%**			**1.50**	**36.3%**

From the 10 tranches: Gini = Σ trapezoid − 1 = 50.3% · AUC = 75.1% · KS = 36.3%. On every individual score (Excel 9_Validation): Gini 51.5% · c-statistic 0.757 · KS 37.4%.

Siddiqi: c ≥ 0.7 is usually adequate but not absolute (p.255); application scorecards typically show KS 40–60% (p.251). Compare with development and out-of-time samples (p.259–261).

Beyond Logistic Regression

Modern ML offers higher accuracy — but at a cost: interpretability and regulatory governance.

ML MODELS IN CREDIT SCORING

Random Forest

✅ Handles non-linearity, missing data; ensemble reduces variance

⚠️ Less interpretable; no native probability calibration

XGBoost / LightGBM

✅ State-of-art accuracy; handles sparse data; native feature importance

⚠️ Complex hyperparameter tuning; prone to overfitting on small samples

Neural Networks

✅ Captures complex interactions; excels with unstructured data (text, images)

⚠️ Black box; extremely high data requirements; slow to explain to regulators

🔴 Note: ML methodology (Random Forest, XGBoost, Neural Networks, SHAP) = Industry Practice — not in Siddiqi (2017, pre-ML era).

EXPLAINABILITY — SHAP VALUES

- SHAP = Shapley Additive exPlanations
- Decomposes each prediction into feature contributions
- Enables 'adverse action' explanations for ML models
- Requirement for high-risk AI (AI Law 2025)

REGULATORY DIMENSION

- AI Law 2025 (134/2025/QH15): credit scoring = high-risk AI system
- Requirements: transparency, explainability, human oversight
- Banks must document model decisions and maintain audit trail
- Regulator requires model validation before deployment

• IP • VN-REG

Block 7: Model Validation

The Scorecard Development Pipeline

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- Correlation checks
- Business logic check

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8 **Reject inference → AGB**

- Final scorecard on AGB
- AI/ML challenger (IP)

OUTPUT

Model Performance Report

- Regression assumptions & results (GINI, KS)
- ML challenger tuning

Validate & Document

9 **Validation & approval**

- Out-of-sample & out-of-time tests
- Model Risk sign-off

OUTPUT

Validation Report

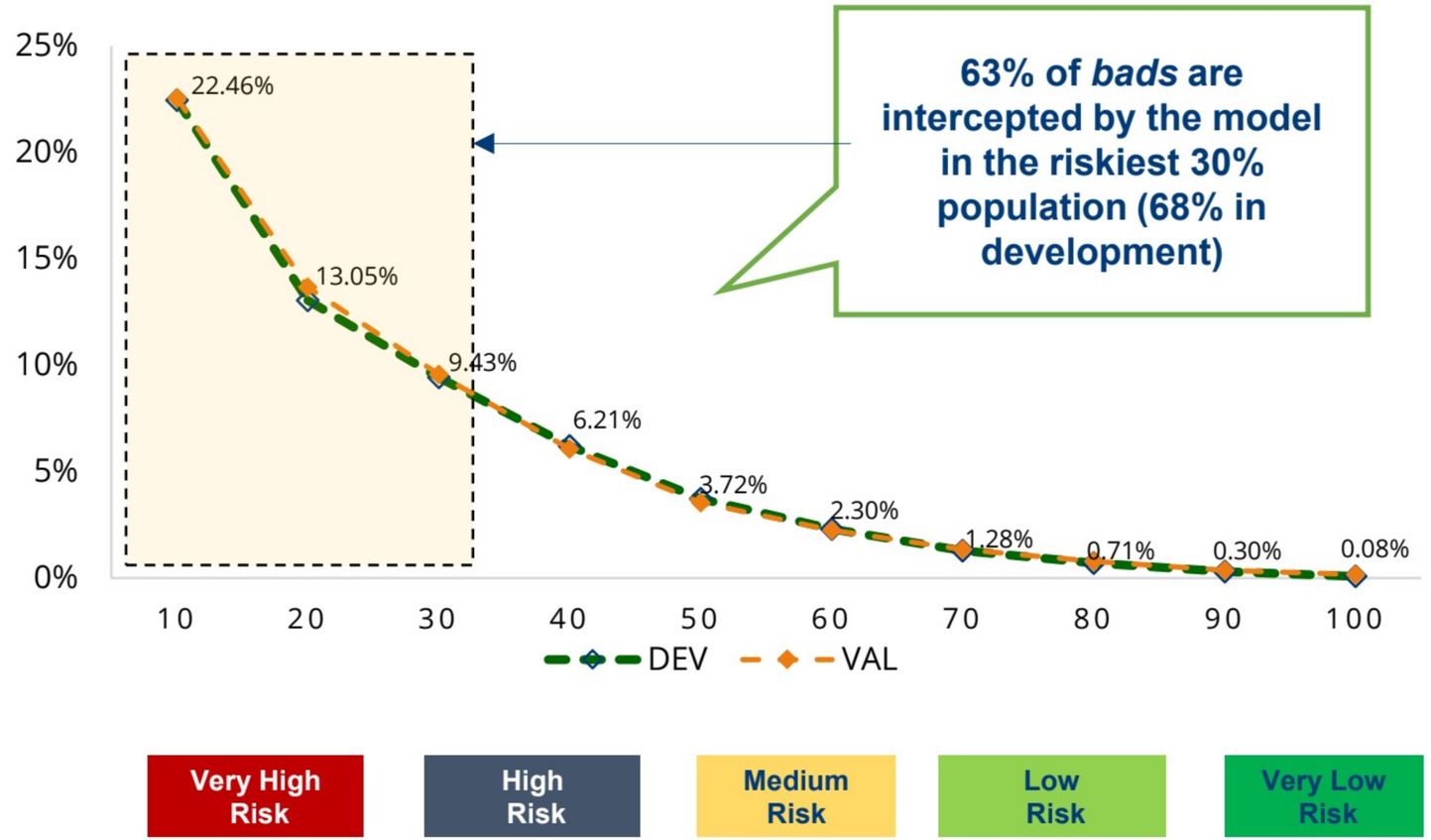
- PSI, CSI (HHI = IP)
- Out-of-time results
- KPI thresholds & full documentation

Iterative: steps 5–7 repeat until the scorecard is stable and makes business sense. **Step 9** needs independent sign-off from Model Risk, not only the model builders.

Model Outcome – Sample Performance

Model Performance

Bad Rate trend by score tranches



Score performance

- GINI Statistic : 52%
- KS Statistic : 37%

